

Homotopy-tilt orbit and spectral collapse

internal tilt orbit in the fixed envelope

$$\mathbb{T}_{\mathcal{R}} = \text{Hom}(F/F_{\text{norm}}, \mathcal{R})$$

$$\dim \mathbb{T}_{\mathcal{R}} = 42$$

$$A_{\tau} = A - \tau q_F D_{\alpha}|_{\mathcal{R}}$$

$$\tau \in \mathbb{T}_{\mathcal{R}}$$

$$\mathcal{O}_{\text{tilt}}(A) = \{B \in \text{End}(\mathcal{R}) : B(E) = -2E\}$$

$$\mathcal{R} = \mathbb{C}E \oplus (\mathcal{R}/\mathbb{C}E)$$

$$B(E) = -2E \text{ fixed}$$

$$\text{quotient operator arbitrary}$$

exact representatives and changing spectra

$$A$$

$$\chi_A(S) = (S+2)^2 q(S)$$

$$B_0 = \text{diag}(-2, 0, \dots, 0)$$

$$\chi_{B_0}(S) = (S+2)S^6$$

$$B_3 = \text{diag}(-2, 3, \dots, 3)$$

$$\chi_{B_3}(S) = (S+2)(S-3)^6$$

all three lie in the same normalized internal tilt orbit

$$\gcd_{\tau} \chi_{A_{\tau}}(S) = S+2$$

same statement for minimal polynomials

$q(S)$, the second -2 line,
and the fixed g_3 -cyclic module
are not invariants